

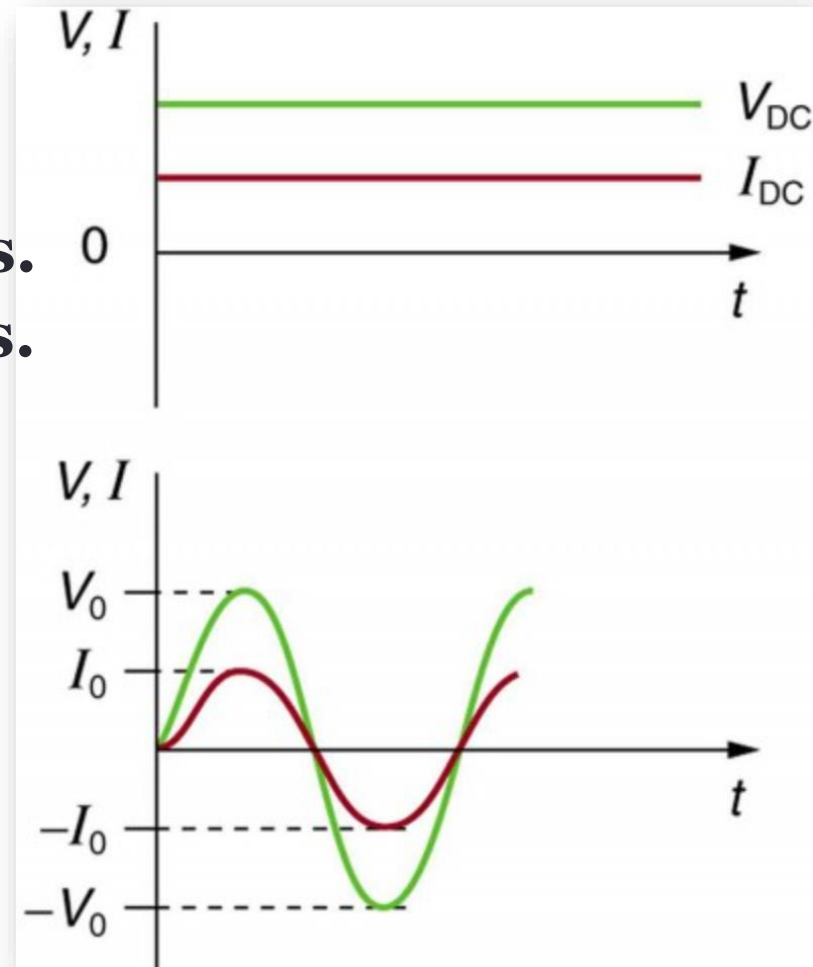
WIDE TUNING RANGE VCOS

SIVARAMAKRISHNA R

Motivation

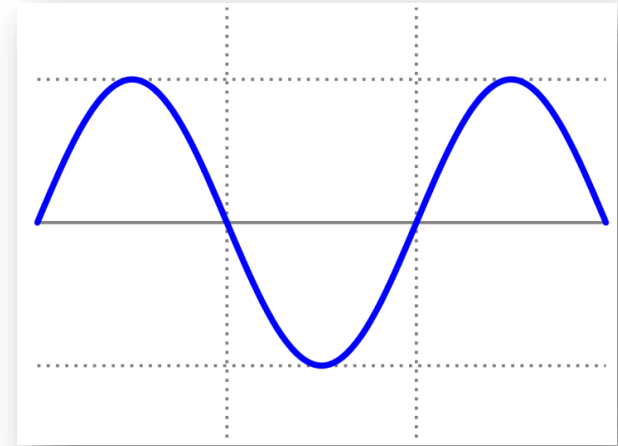
Just like the way we need DC power supply
We also need (periodically) time vary signals

- ☐ **Wireline Communication Sys.**
- ☐ **Wireless Communication Sys.**
 - ☐ **Up Conversion**
 - ☐ **Down Conversion**
 - ☐ **Provides the carrier frequency**
- ☐ **Radars**



Performance Metrics

- ❑ DC power dissipation
- ❑ Amplitude
- ❑ Tunability
- ❑ Phase noise
- ❑ Quadrature Phases



Performance Metrics

□ Tunability

- Wide tuning range is desired

2G

GSM band ↕	f (MHz) ↕	Uplink (MHz) (mobile to base) ↕	Downlink (MHz) (base to mobile) ↕
T-GSM-380 ^[a]	380	380.2 – 389.8	390.2 – 399.8
T-GSM-410 ^[a]	410	410.2 – 419.8	420.2 – 429.8
GSM-450	450	450.6 – 457.6	460.6 – 467.6
GSM-480	480	479.0 – 486.0	489.0 – 496.0
GSM-710	710	698.2 – 716.2	728.2 – 746.2
GSM-750	750	777.2 – 792.2	747.2 – 762.2
T-GSM-810 ^[a]	810	806.2 – 821.2	851.2 – 866.2
GSM-850	850	824.2 – 848.8	869.2 – 893.8
P-GSM-900 ^[d]	900	890.0 – 915.0	935.0 – 960.0
E-GSM-900 ^[e]	900	880.0 – 915.0	925.0 – 960.0
R-GSM-900 ^[h]	900	876.0 – 915.0	921.0 – 960.0
T-GSM-900 ^[a]	900	870.4 – 876.0	915.4 – 921.0
DCS-1800 ^[i]	1800	1710.2 – 1784.8	1805.2 – 1879.8
PCS-1900 ^[j]	1900	1850.2 – 1909.8	1930.2 – 1989.8

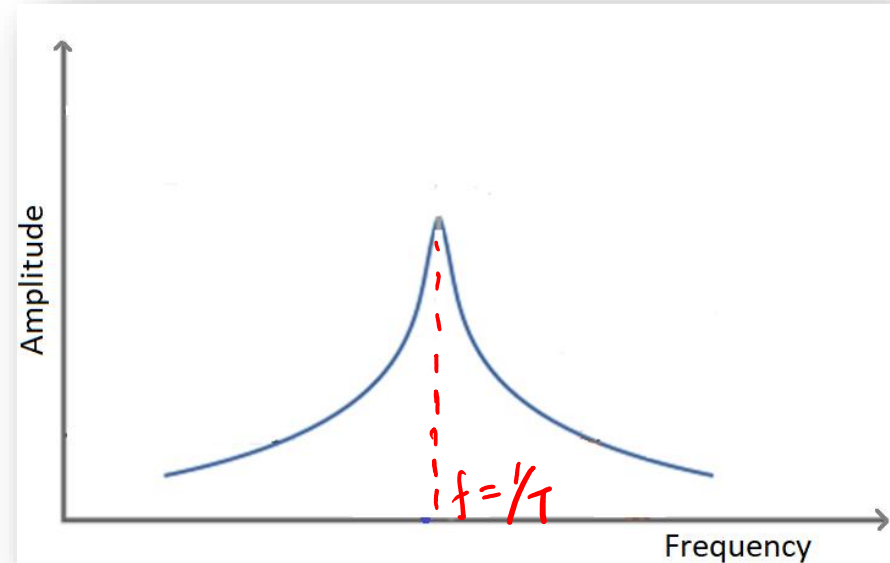
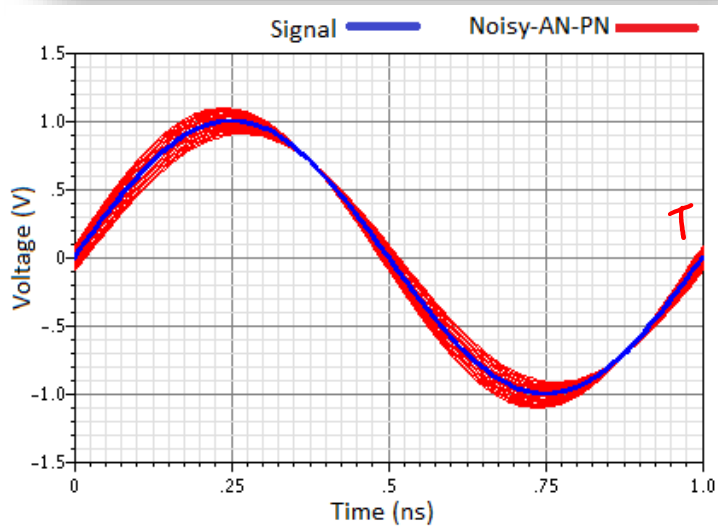
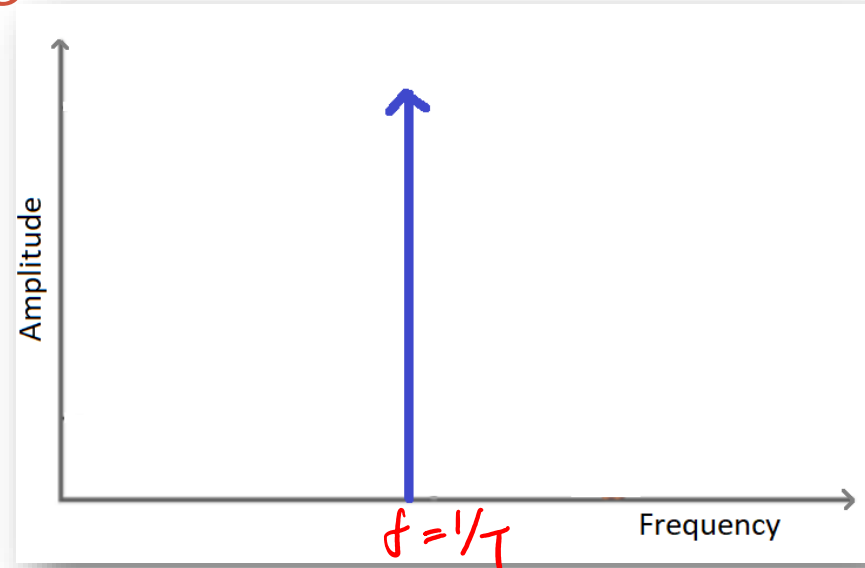
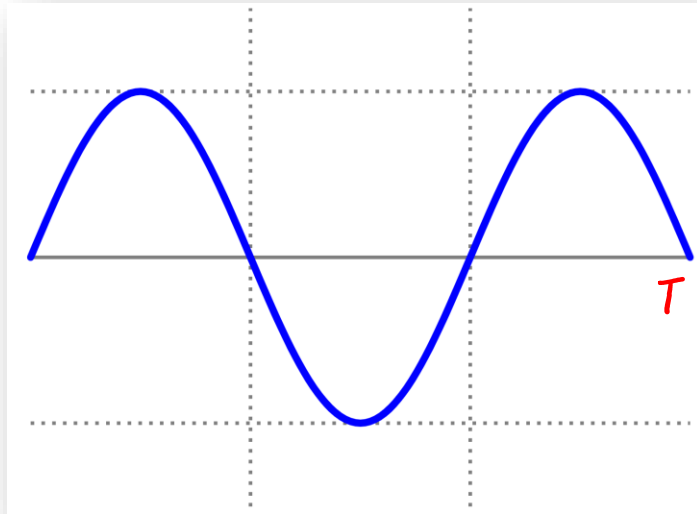
4G

Band ↕	Duplex mode ^[A 1] ↕	f (MHz) ▲
87	FDD	410
88	FDD	410
31	FDD	450
72	FDD	450
73	FDD	450
71	FDD	600
12	FDD	700

30	FDD	2300
40	TDD	2300
41	TDD	2500
53	TDD	2500
7	FDD	2600
38	TDD	2600

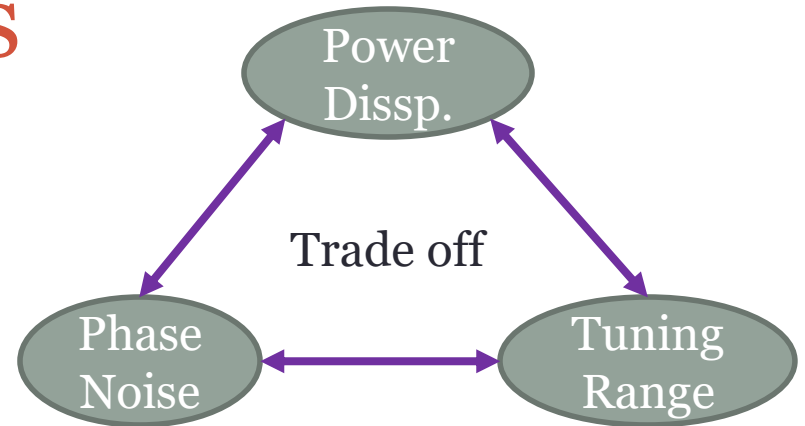
Performance Metrics

Phase noise - Jitter



Performance Metrics

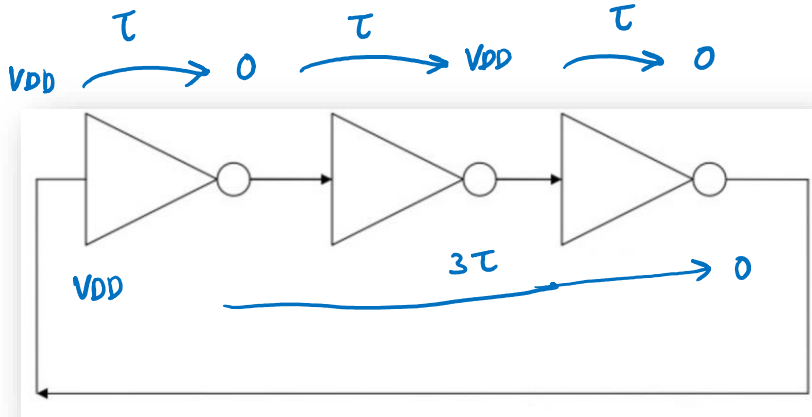
- ❑ DC power dissipation
- ❑ Amplitude
- ❑ Tunability
- ❑ Phase noise
- ❑ Quadrature Phases



$$FOMT = 20 \log \left(\frac{f_o}{\Delta f} \frac{FTR}{10} \right) - PNoise(\Delta f) |_{dB} - 10 \log(P_{DC})$$

$$FTR = 200 \frac{f_{\max} - f_{\min}}{f_{\max} + f_{\min}}$$

Oscillators

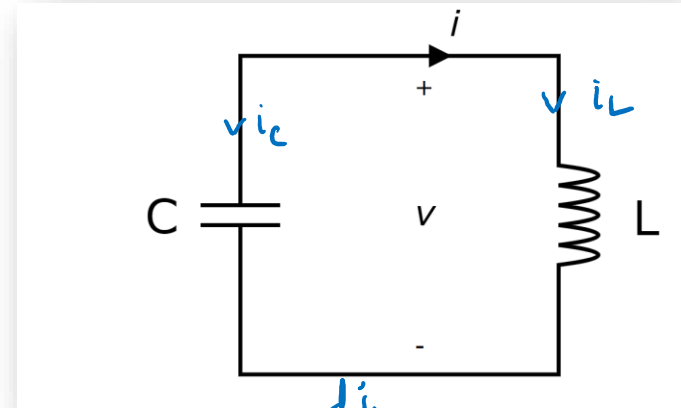


$$\text{Time period} = (3\tau) \times 2 = 6\tau$$

$$\Rightarrow f = \frac{1}{6\tau}$$

$$\text{In general: } f = \frac{1}{2NT}$$

Square wave



$$v = L \frac{di_L}{dt}$$

$$-i_L = i_c = C \frac{dv}{dt}$$

$$\Rightarrow v = -LC \frac{dv}{dt}$$

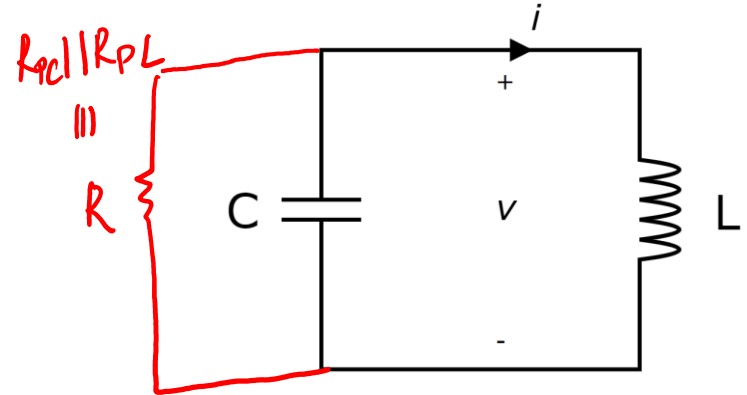
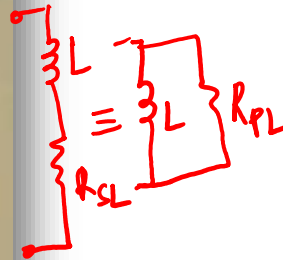
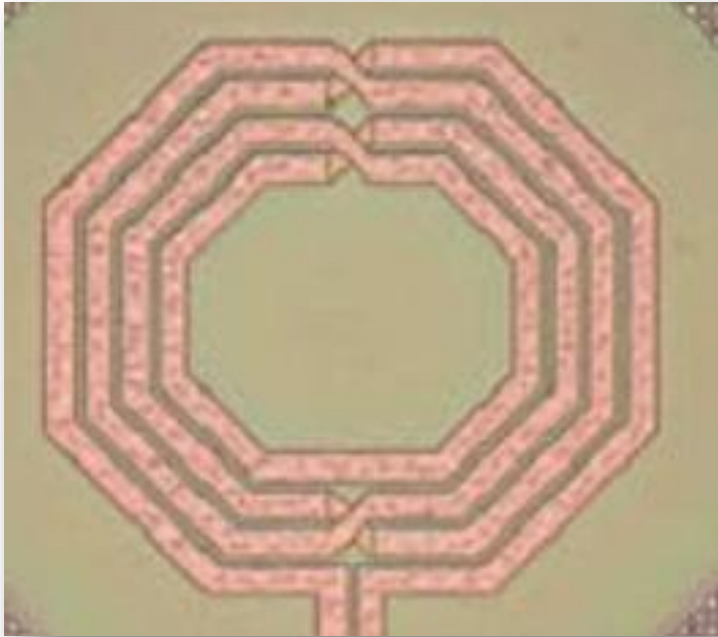
$$v(t) = V_0 \cos[\omega_0 t + \phi]$$

Sinusoidal

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$L = 2 \text{ nH}, C = 12 \text{ pF} \rightarrow f \approx \underline{\underline{16}}$$

Oscillators



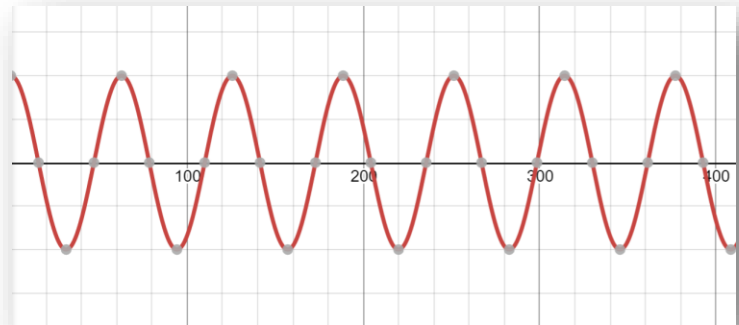
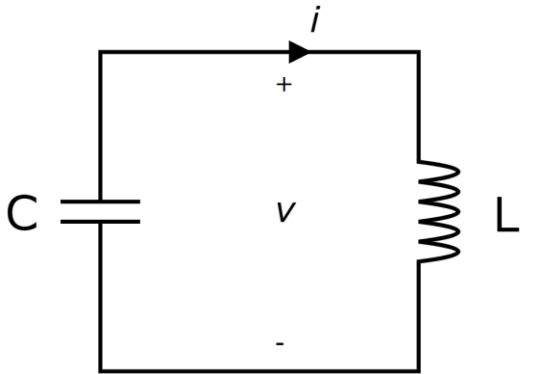
Non-ideal components;
Quality factor



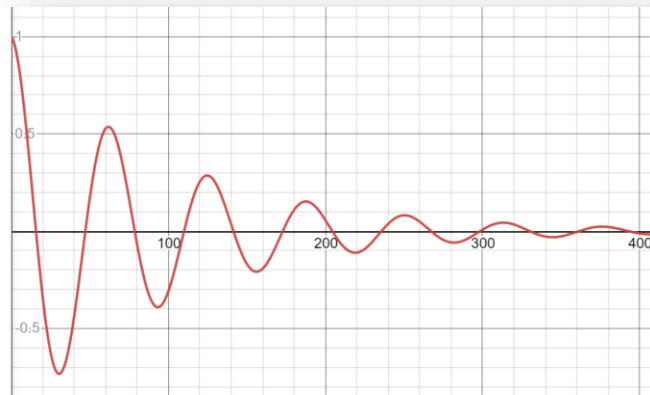
$$v(t) \simeq V_0 e^{-t/RC} \cos(\omega t)$$

Oscillators

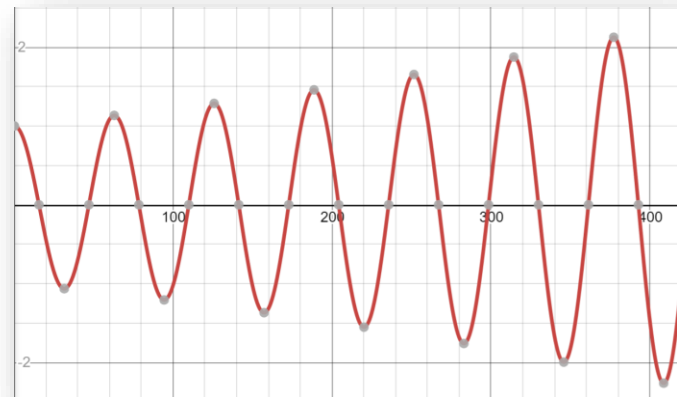
$$v(t) = V_0 e^{-t/RC} \cos(\omega t) \quad R = \text{Infinite!}$$



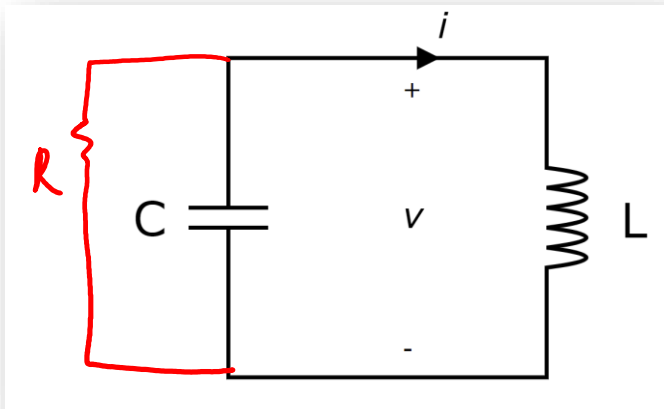
R is positive



R is negative

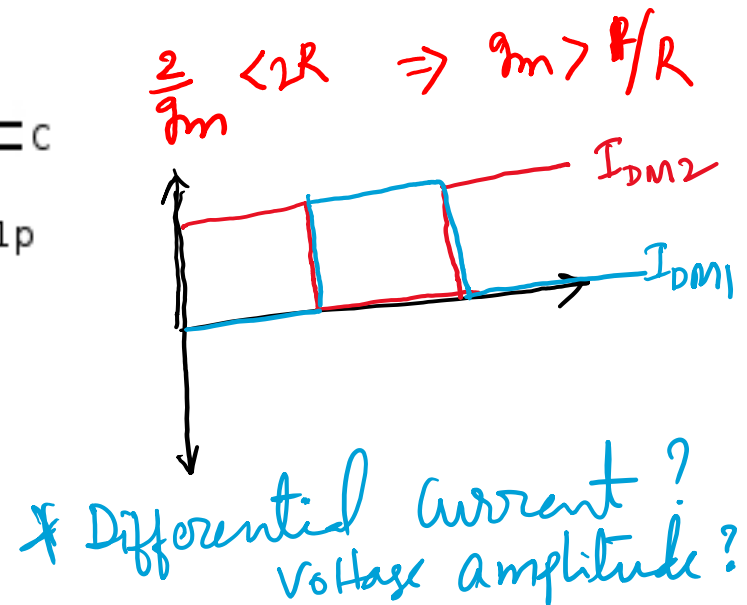
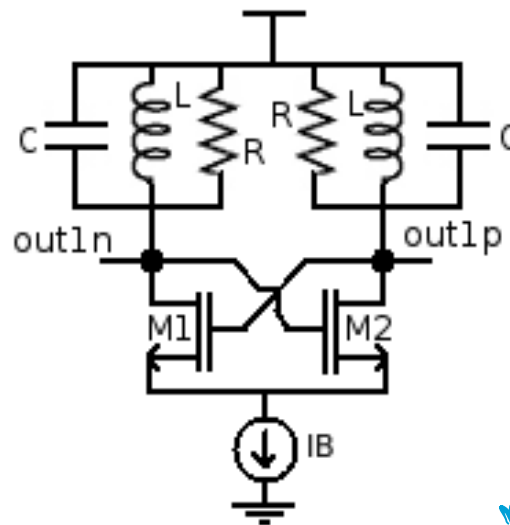
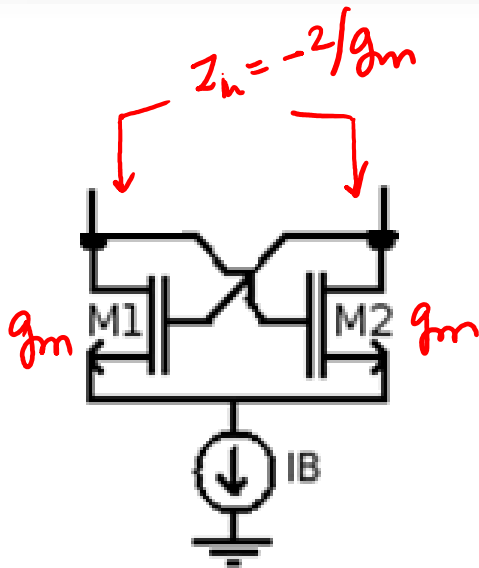


LC Oscillators



$$R_{eff} = \frac{-R_0 R}{-R_0 + R}$$

R_{eff} is negative if $R_0 < R$ (i.e, $|R_0| < |R|$)

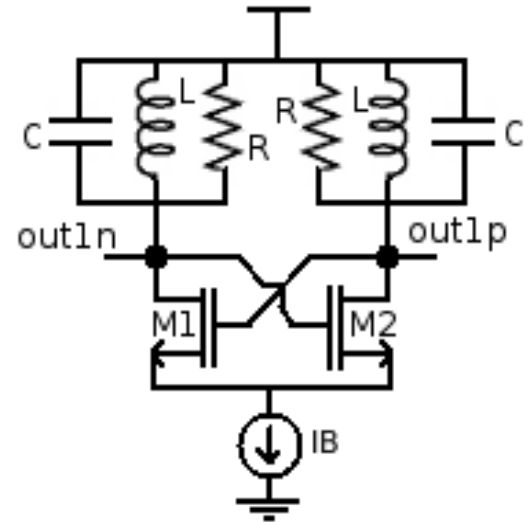


LC Oscillator

- Amplitude estimation :

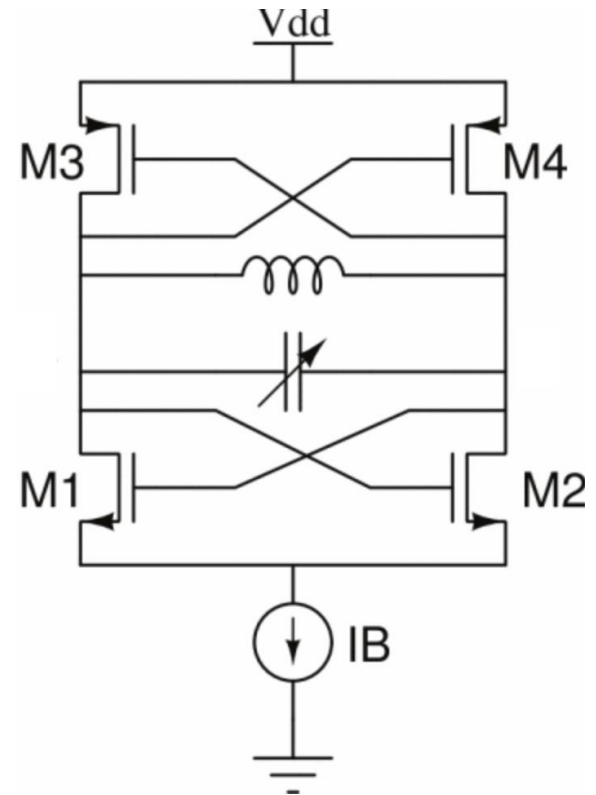
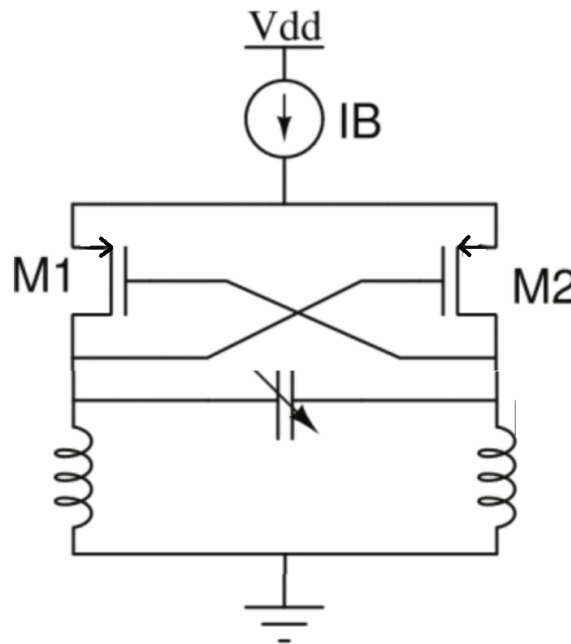
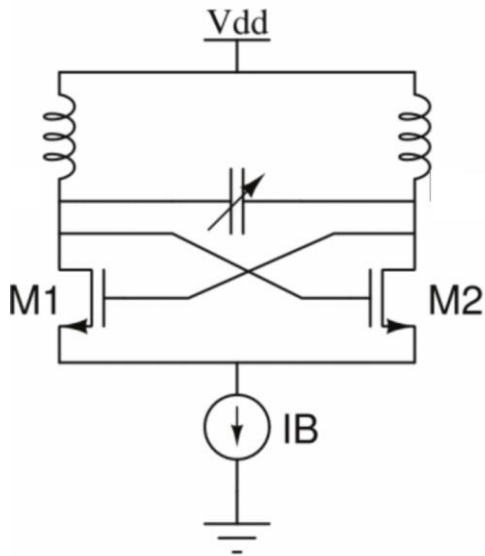
$$V_o = \frac{4}{\Pi} * \frac{IB}{2} * 2 * R$$
$$= \frac{4}{\Pi} * IB * R$$

- Amplitude increases with current
 - Up to a certain point: Voltage limited regime
- Amplitude increases with current

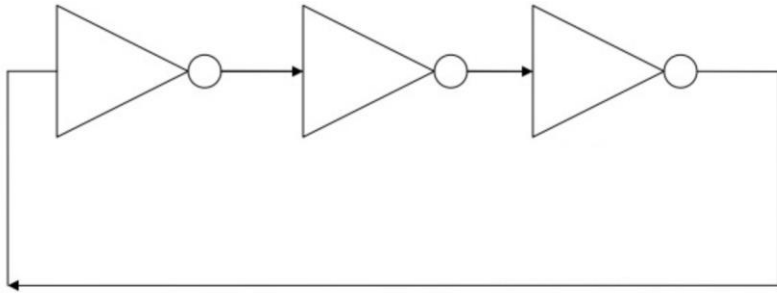


LC Oscillators

- NMOS, PMOS or CMOS
- Trade-off: Power vs. phase noise

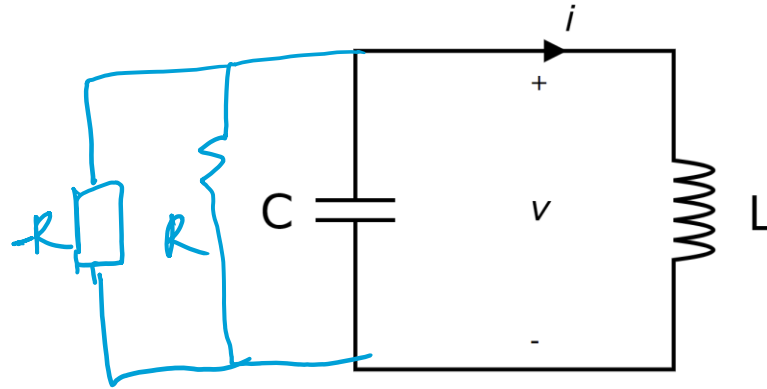


Oscillators



Ring Oscillators

- Less power dissipation
- Poor phase noise



LC Oscillators

- More power dissipation
- Superior phase noise

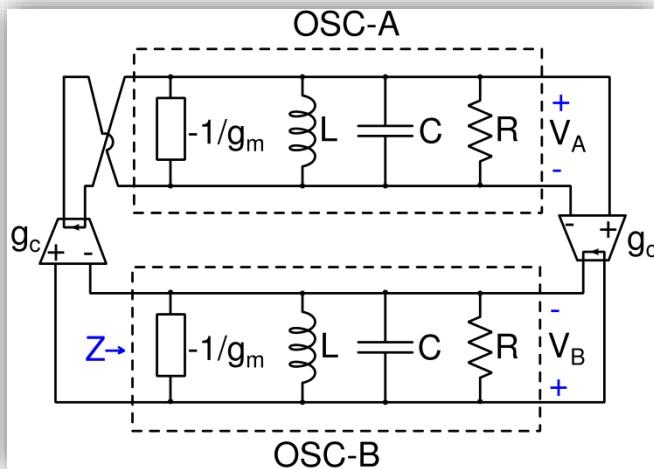
Tunability in Oscillators

- Variable Capacitor or Variable Inductor

Quadrature Phase Generation

- Using frequency dividers
- However, we need CLK to be twice the (carrier) frequency
- Quadrature phase LC oscillators

Quadrature Phase LC-Oscillator: Linear Model



$$V_A = \pm jV_B$$

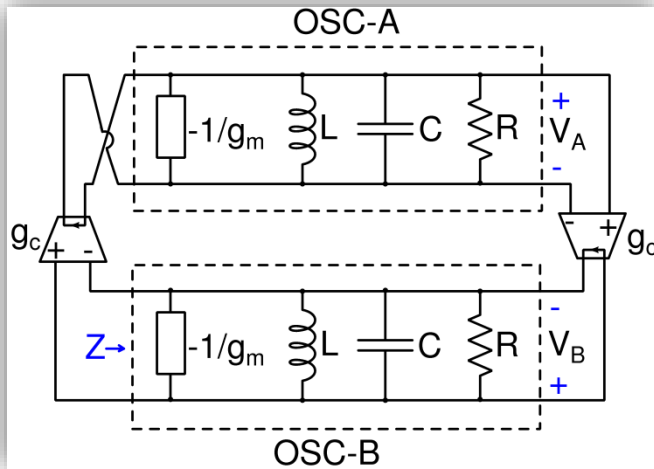
$$\omega = \sqrt{\frac{g_c^2 L^2 + 4LC}{2LC}} \pm \frac{g_c}{2C}$$

$$\approx \omega_0 \pm \frac{g_c}{2C} \quad \text{where } \omega_0 = \frac{1}{\sqrt{LC}}$$

Loop analysis:

- Two possible frequencies of operation
- Two possible phase sequences
- How to resolve ambiguities in frequency of operation and phase sequence

Quadrature Phase LC-Oscillator: Linear Model



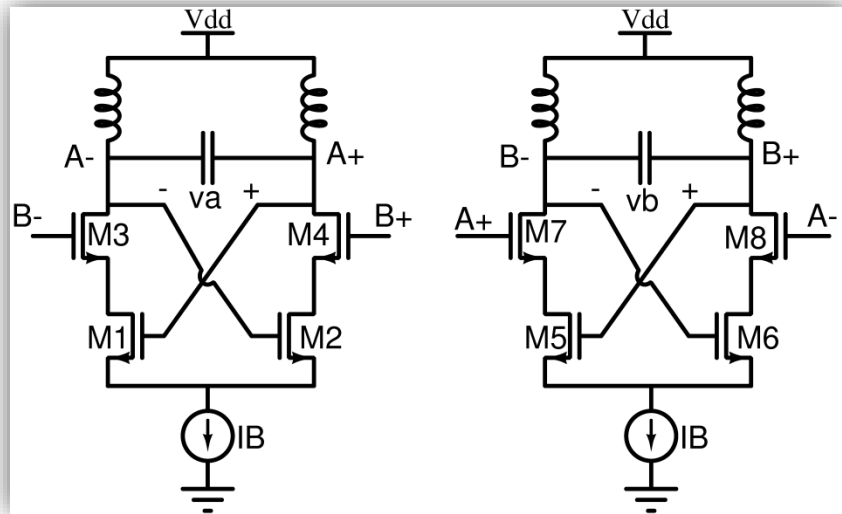
$$\omega = \begin{cases} \sqrt{\frac{g_c^2 L^2 + 4LC}{2LC}} + \frac{g_c}{2C} & \text{when B leads A} \\ \sqrt{\frac{g_c^2 L^2 + 4LC}{2LC}} - \frac{g_c}{2C} & \text{when B lags A} \end{cases}$$

$$\Rightarrow \omega = \begin{cases} \omega_0 + \frac{g_c}{2C} & \text{when B leads A} \\ \omega_0 - \frac{g_c}{2C} & \text{when B lags A} \end{cases} \quad \text{where } \omega_0 = \frac{1}{\sqrt{LC}}$$

Link between oscillation frequency and phase sequence is established!

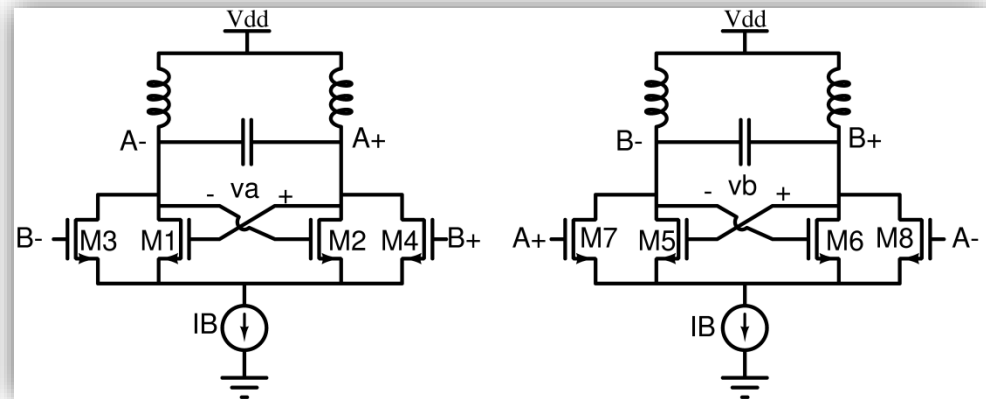
- QVCO drives into a mode in which it sees more $-g_{\text{eff}}$ (due to active circuitry and/or LC tank)
- Frequency can be tuned by varying g_c
 - Phase sequence should be maintained
 - Frequency tuning can be further enhanced from $\omega_0 - |g_{c,\text{max}}|/2C$ to $\omega_0 + |g_{c,\text{max}}|/2C$ (by switching the polarity of g_c without dissipating extra power)

Coupling Topologies in LC-QVCOs



Series coupling

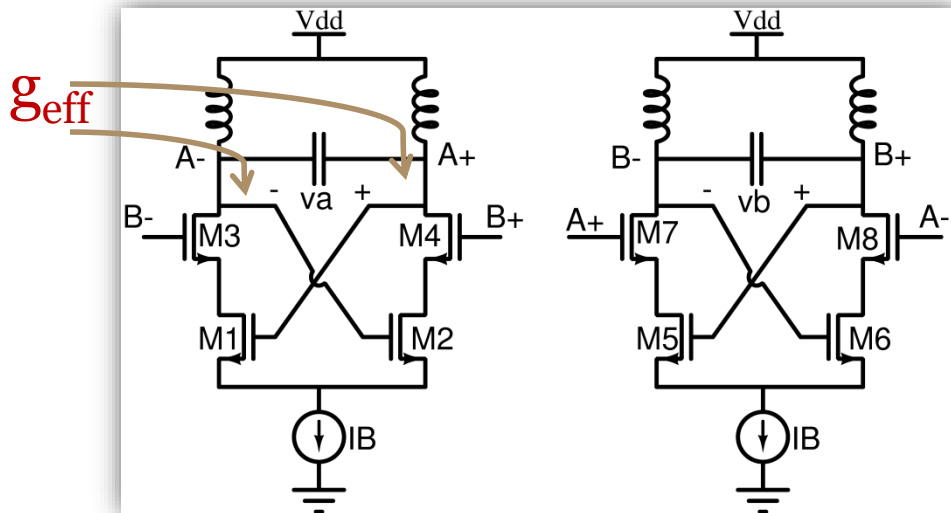
- Coupling network is in series
- Coupling strength is constant



Parallel coupling

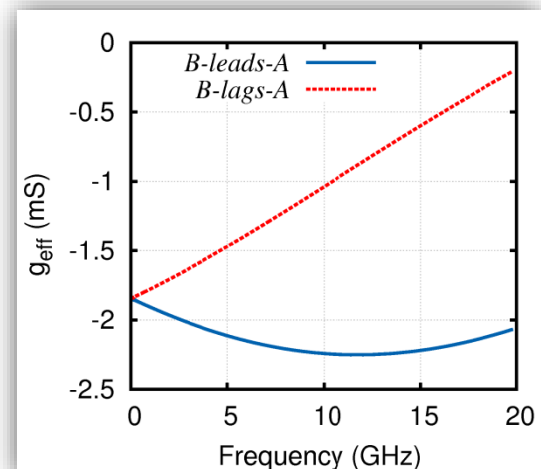
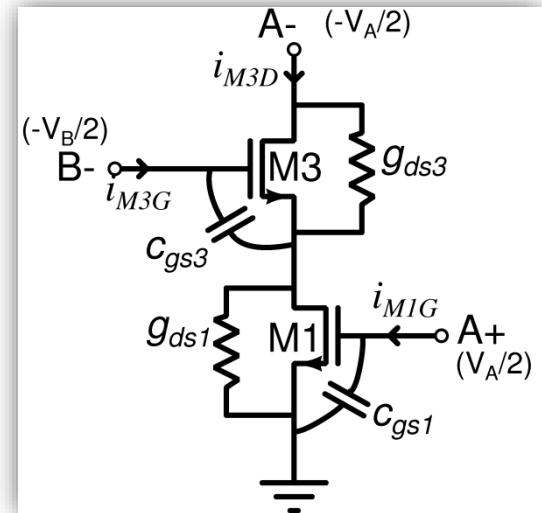
- Coupling network is in parallel
- Coupling strength can be varied

Coupling Topologies: Series Coupling

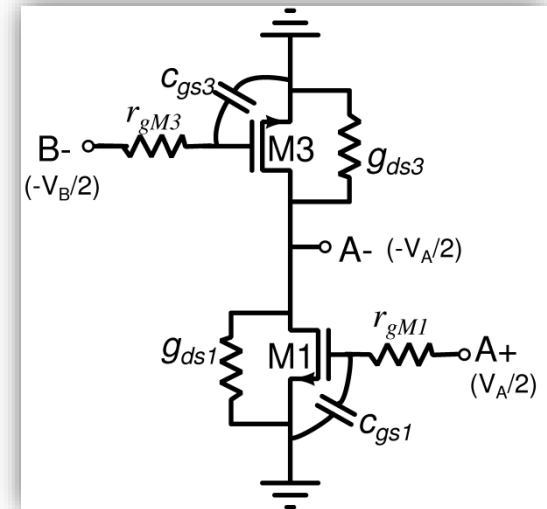
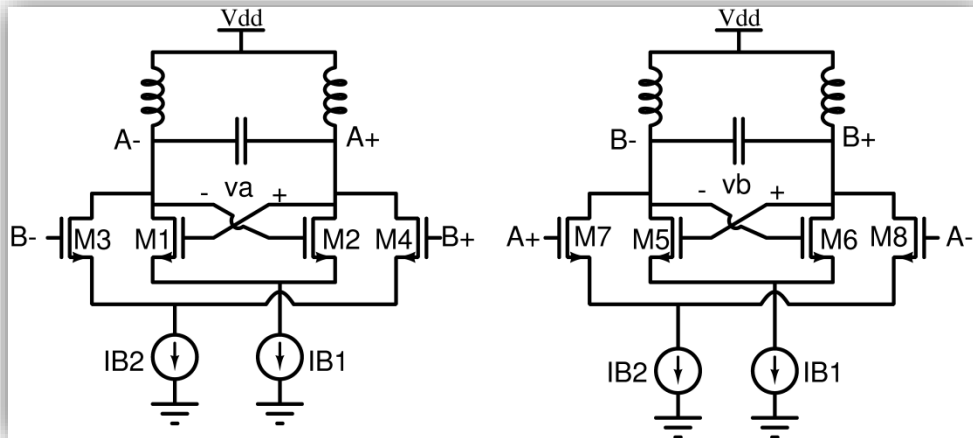


$$g_{eff} = -\frac{1}{2} \left[\frac{g_{m1}g_{m3}^2 + g_{m1}g_{m3}g_{ds1} - g_{ds1}\omega^2c_{gs3}^2}{(g_{m3} + g_{ds1})^2 + \omega^2c_{gs3}^2} \right. \\ \left. \pm \frac{g_{m1}g_{m3}\omega c_{gs3} - g_{m3}g_{ds1}\omega c_{gs3} + g_{m1}g_{ds1}\omega c_{gs3}}{(g_{m3} + g_{ds1})^2 + \omega^2c_{gs3}^2} \right]$$

- Small signal analysis still hold during the startup
- Positive sign for *B-leads-A* & negative sign for *B-lags-A*
- Series coupling favours *B-leads-A* mode

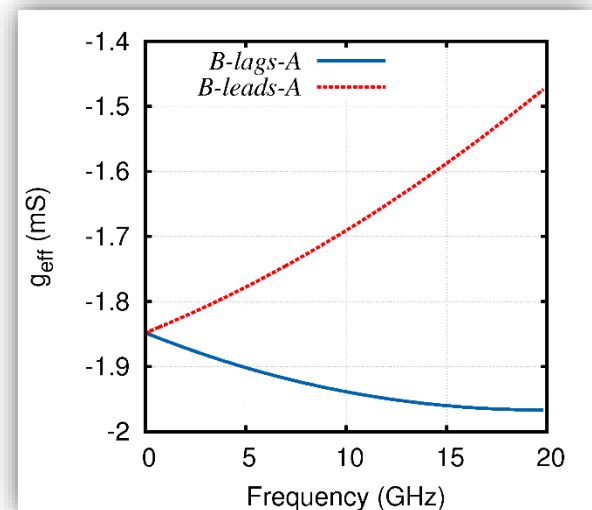


Coupling Topologies: Parallel Coupling

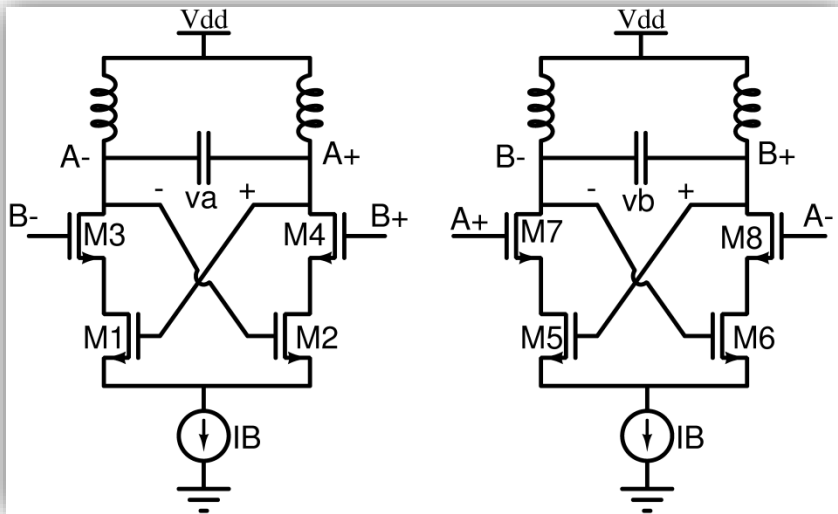


$$g_{eff} = -\frac{1}{2} \left[g_{m1} - \omega^2 c_{gs1}^2 r_{g1} - \omega^2 c_{gs3}^2 r_{g3} \mp g_{m3} \omega c_{gs3} r_{g3} \right]$$

- Difference in g_{eff} is not that significant for *B-leads-A* and *B-lags-A* modes
- r_g can play a role at mm-Wave frequencies
- Coupling strength can be varied with $IB2$
- Frequency can be tuned

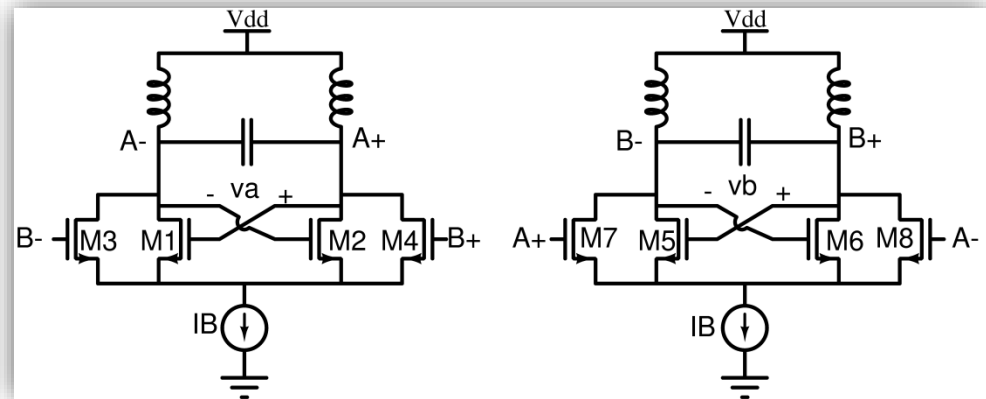


Coupling Topologies in LC-QVCOs



Series coupling

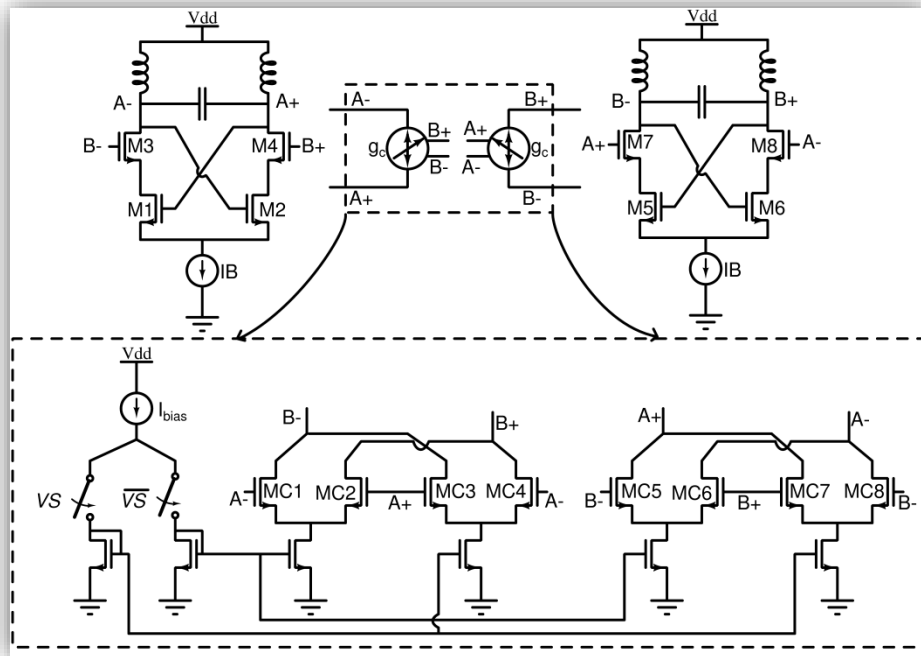
Can be used for deterministic phase sequence



Parallel coupling

Can be used for tuning the frequency

Proposed Series Parallel Coupling LC-QVCO



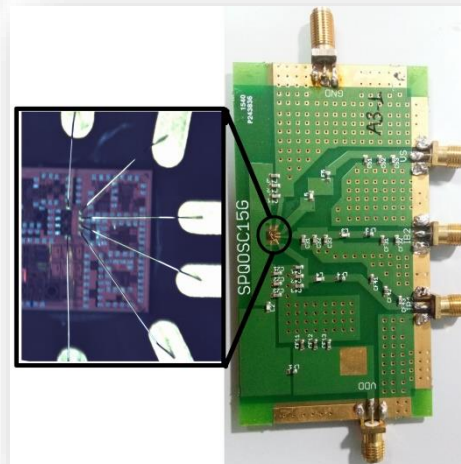
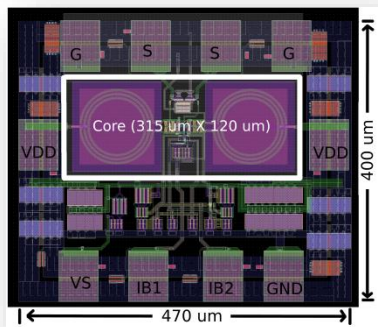
Series coupling: deterministic phase sequence

Parallel coupling: tunability

g_c is determined by I_{bias}

Current steering mechanism, based on VS , to switch the polarity of g_c

$\omega_o - |g_{c,max}|/2C$ to $\omega_o + |g_{c,max}|/2C$

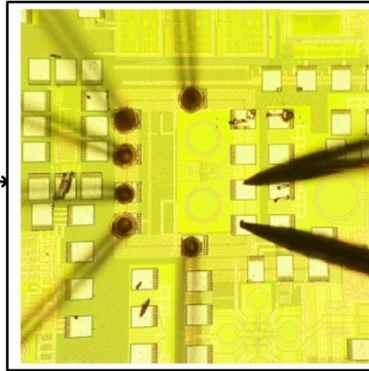


Proposed Series Parallel Coupling LC-QVCO



Keithley SMUs

DC Connections
(VDD, GND, IBs)

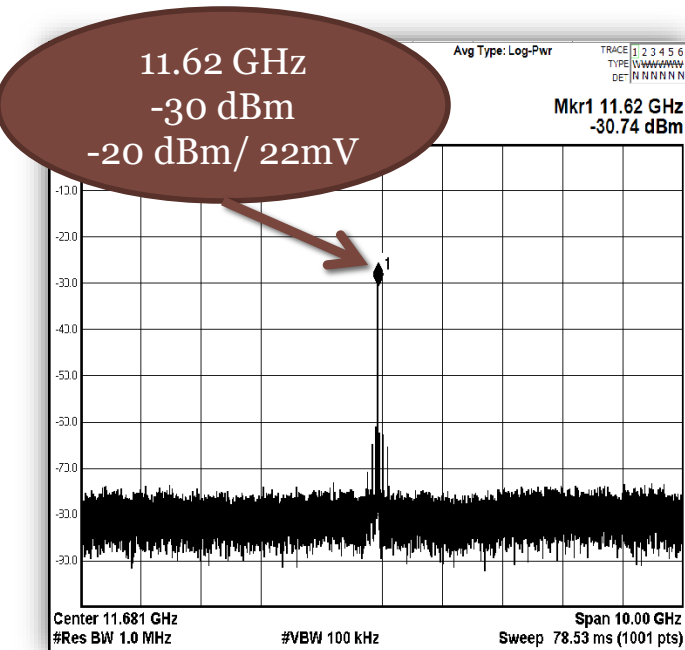


DUT

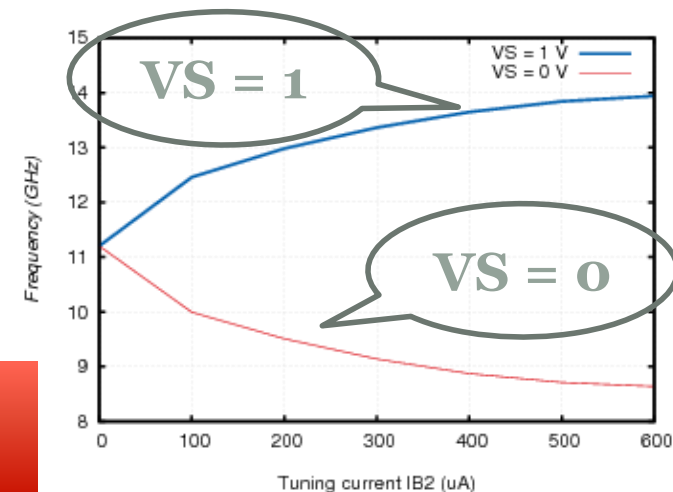
GGB RF Probes
(RF_{out})



R&S FSV30

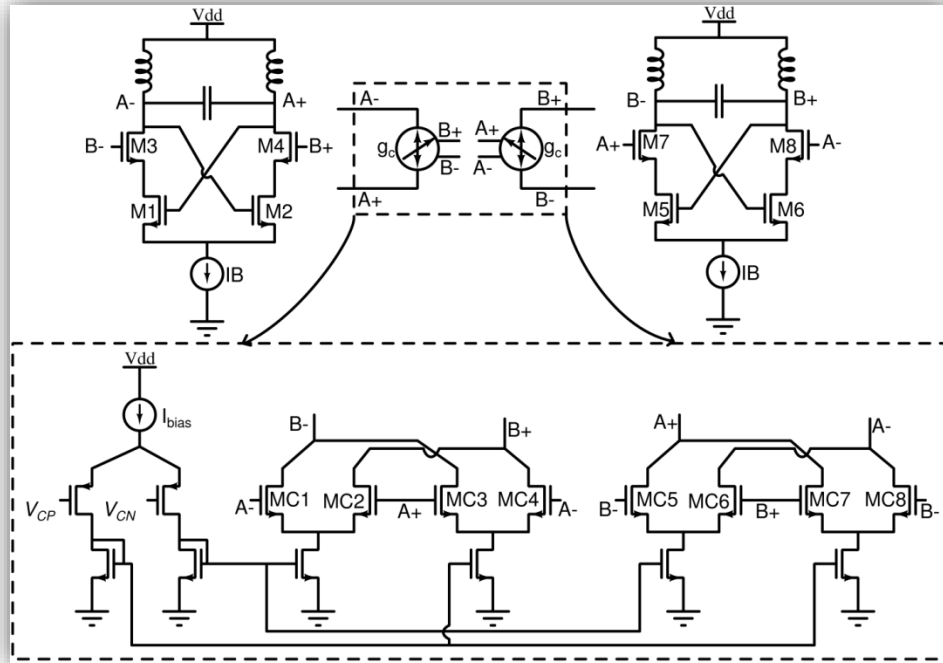


Single ended output
Cable losses



- Poor phase noise ($-69 \leftrightarrow -72$ dBc/Hz @ 1MHz offset)
- Switching between current paths around the center frequency

SPQVCO: “Differential” Parallel Coupling

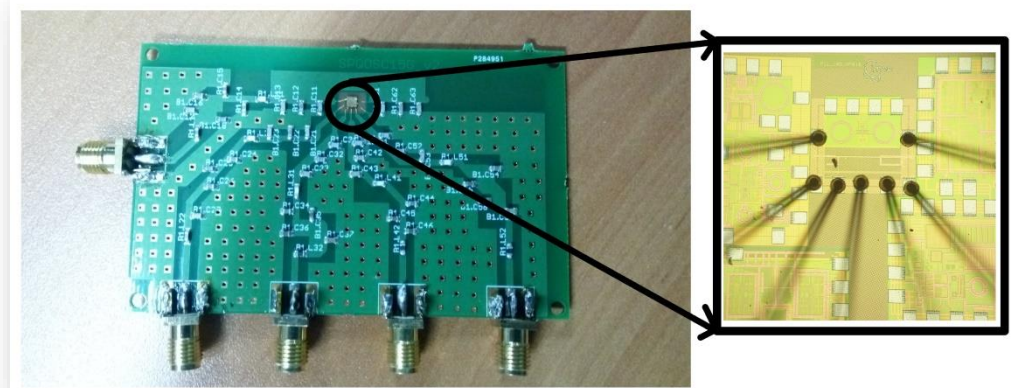
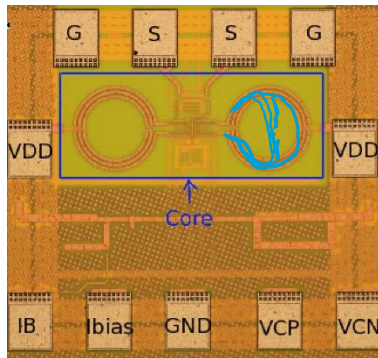


No ‘hard’ switching of parallel coupling

Single sweep

‘Dynamic’ tuning range

Core area : 320 X 150 μm^2



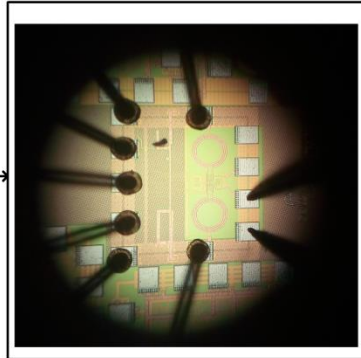
Proposed Series Parallel Coupling LC-QVCO

Measurements : Die on Board!



Keithley SMUs

DC Connections
(VDD, GND, IBs)



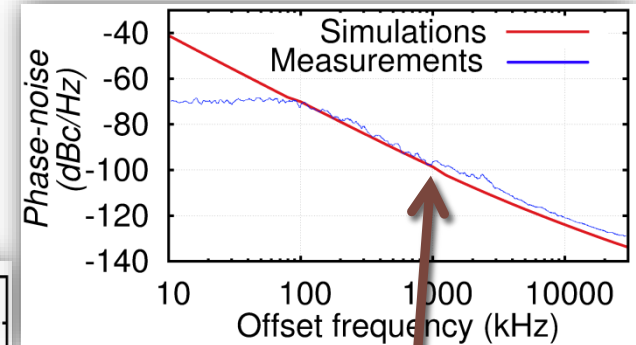
DUT

GGB RF Probes
(RF_{out})

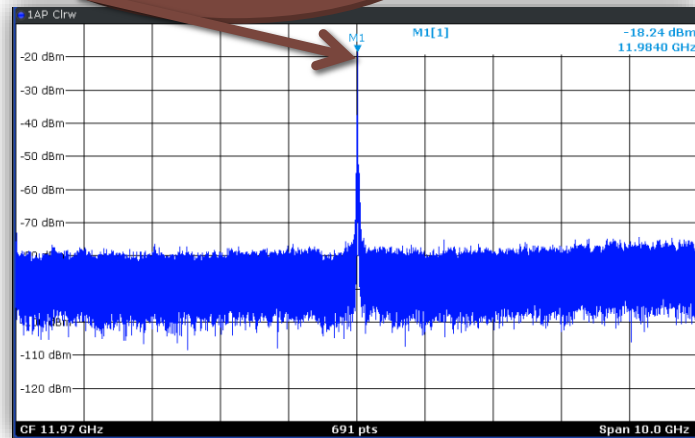
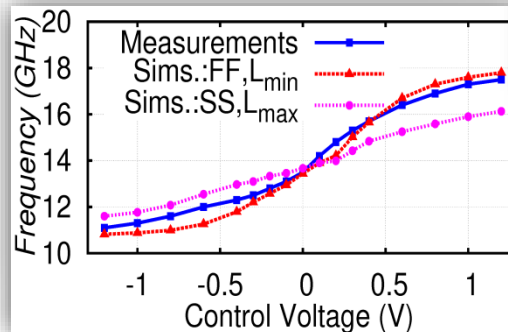


R&S FSV30

11.62 GHz
-18.5 dBm
-8.5 dBm / 120 mV



-96.9 dBc/Hz
@1MHz



Proposed SPQVCOs

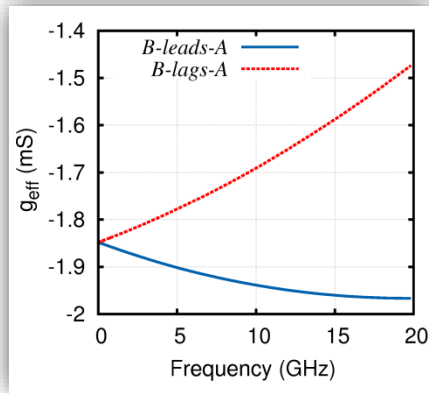
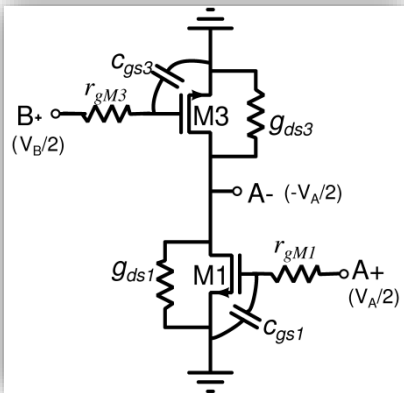
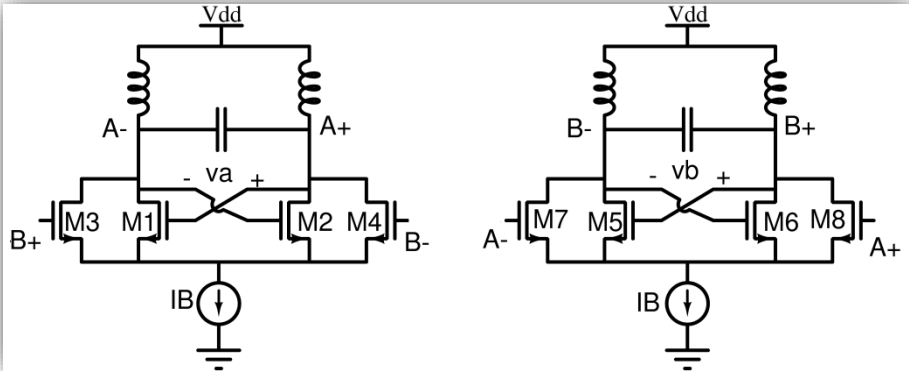
Comparison with existing designs

Ref.	Tech.	f_{center} (GHz)	FTR (%)	FOMT (dB)	Remarks
SiRF'11	90 nm	13.1	21.4	186.7	IC
JSSC'11	90nm	15.1	68.8	186	Variable L
CICC'11	180nm	26.25	14.8	188	--
TCAS-II'12	130 nm	14.1	13.3	188.3	--
EuMIC'13	130 nm	22.8	13.0	170.1	--
This Work	65 nm	14.3	44.7	185	--

$$FTR = 200 \frac{f_{\max} - f_{\min}}{f_{\max} + f_{\min}}$$

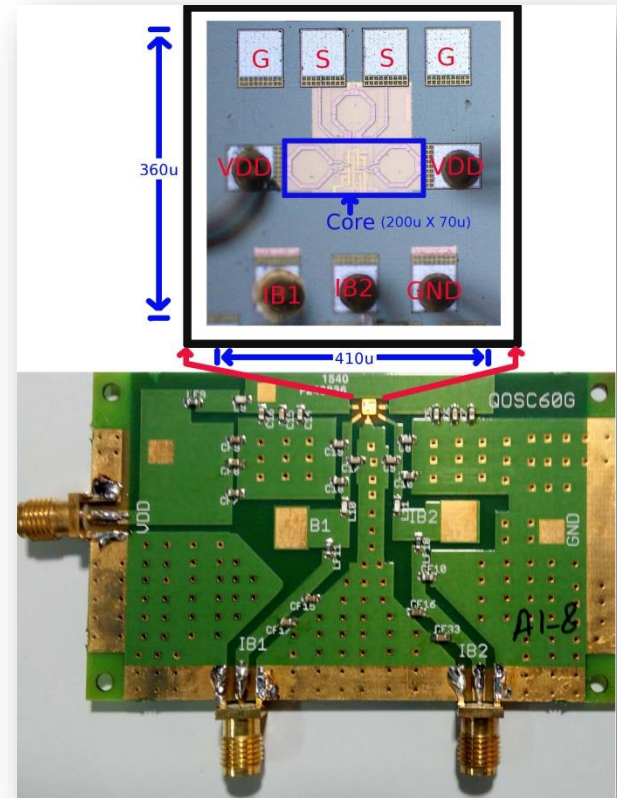
$$FOMT = 20 \log \left(\frac{f_o}{\Delta f} \frac{FTR}{10} \right) - P_{\text{Noise}}(\Delta f) |_{\text{dB}} - 10 \log(P_{DC})$$

mm-Wave PQVCO



$$g_{eff} = -\frac{1}{2} \left[g_{m1} - \omega^2 c_{gs1}^2 r_{g1} - \omega^2 c_{gs3}^2 r_{g3} \pm g_{m3} \omega c_{gs3} r_{g3} \right]$$

Different g_{eff} for lead-lag modes at mm-Wave frequencies



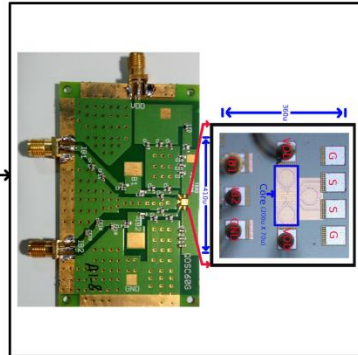
A-leads-B
is dominant mode

mm-Wave PQVCO: Experimental Results



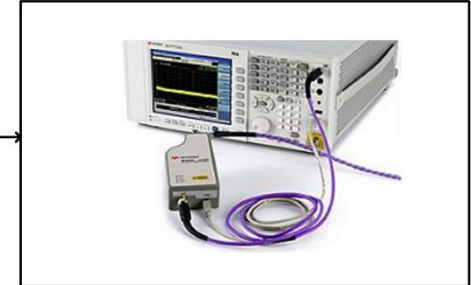
Keithley SMUs

DC Connections
(VDD, GND, IBs)

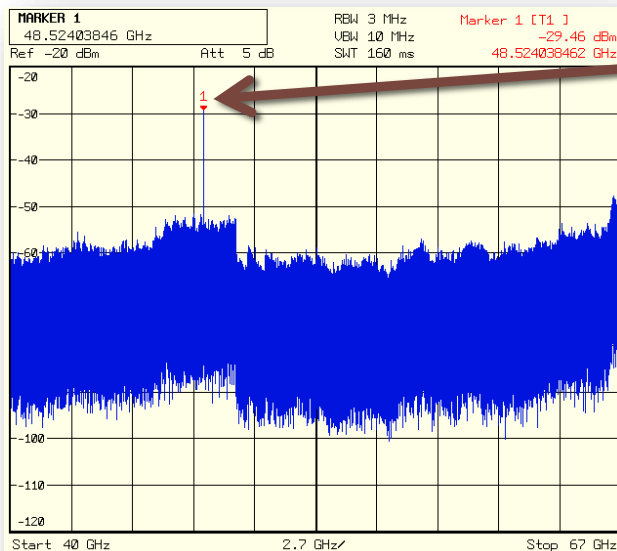


DUT

GGB RF Probes
(RF_{out})



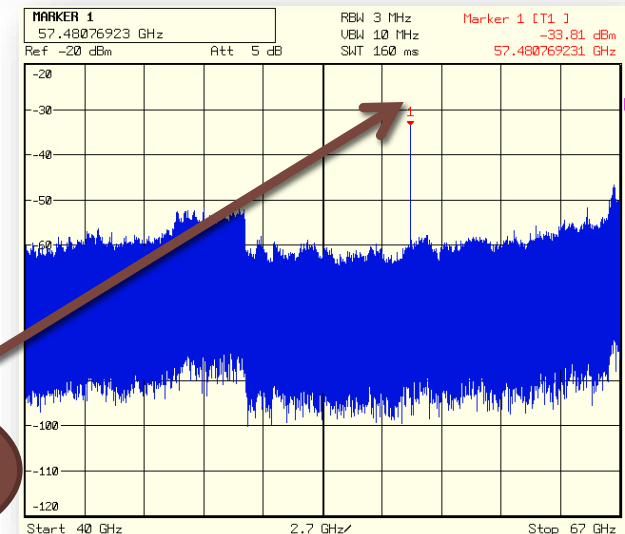
Keysight N9030B Signal Analyzer with
M1970V (external subharmonic) mixer



48.5 GHz
-29 dBm
-19 dBm / 36 mV

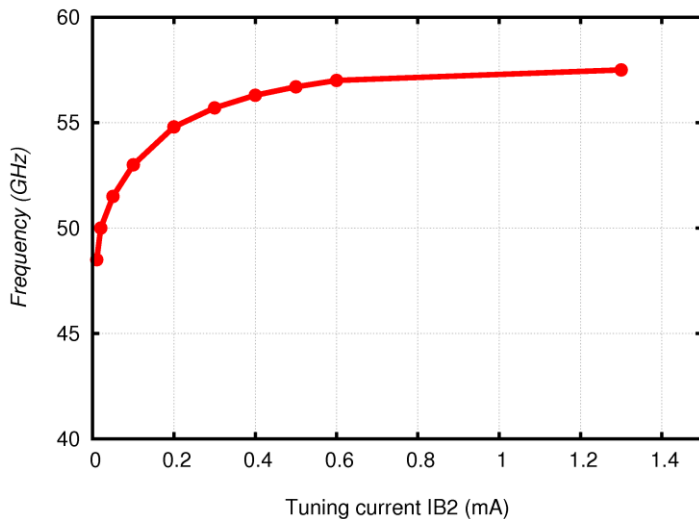
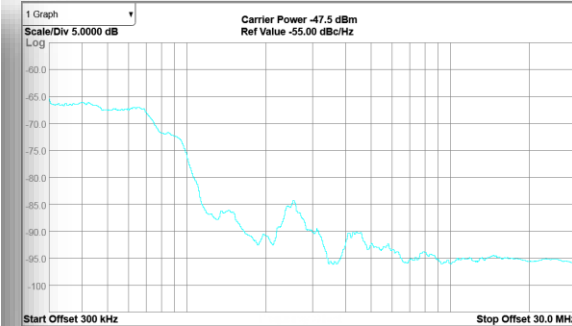
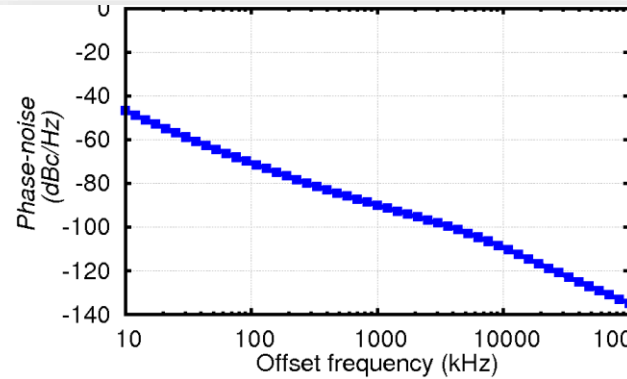
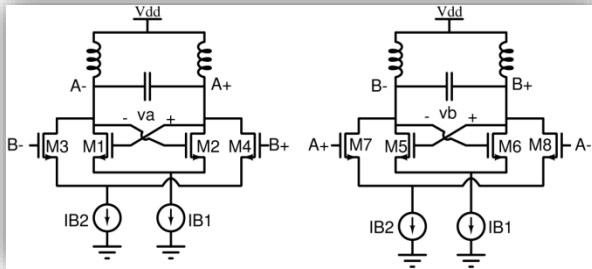
- Single ended
- Cable losses

57.5 GHz
-33.8 dBm
-24 dBm / 20 mV



ACK: Measurements have been carried out at SAC, ISRO, Ahmedabad

mm-Wave PQVCO: Performance Metrics



Technology

UMC 90nm

TR

48.5 – 57.5 (GHz)

PhaseNoise 1@MHz

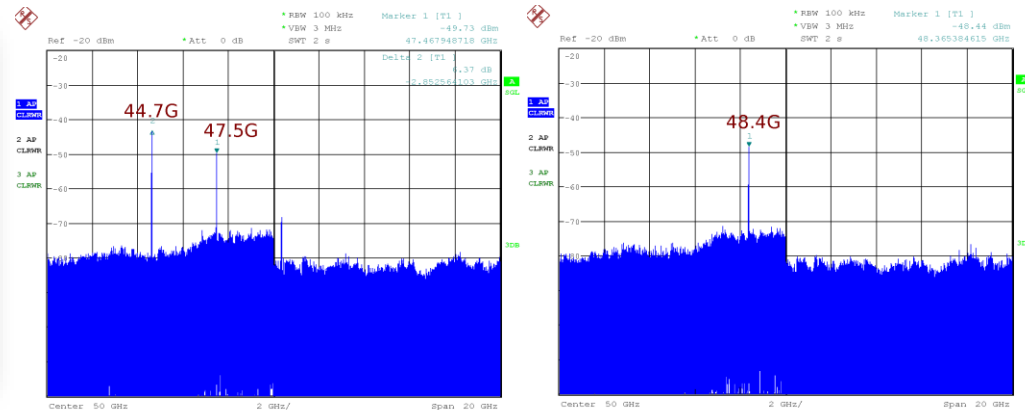
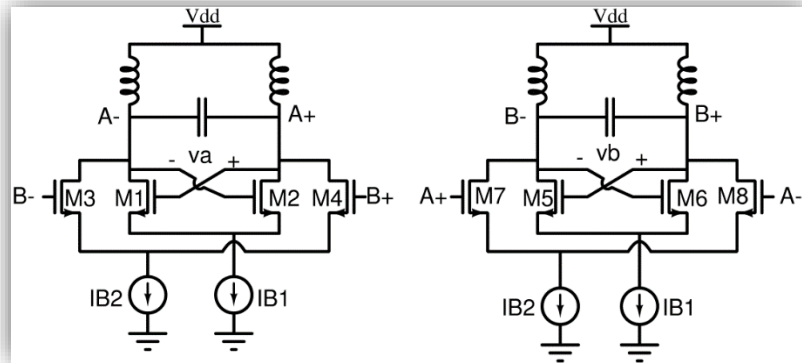
~ -80 dBc/Hz

FOM_T

169

$$FOM_T = 20 \log \left(\frac{f_o}{\Delta f} \frac{FTR}{10} \right) - PNoise(\Delta f)_{dB} - 10 \log(P_{DC})$$

mm-Wave PQVCO: Experimental Results



$$\omega = \begin{cases} \omega_0 + \frac{g_c}{2C} & \text{when B leads A} \\ \omega_0 - \frac{g_c}{2C} & \text{when B lags A} \end{cases} \quad \text{where } \omega_0 = \frac{1}{\sqrt{LC}}$$

- Weak coupling can drive VCO in to both lead-lag modes
- Irrespective of initial conditions, QVCO found to go in to the “**dominating mode**”

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THANK YOU
